
Data With the Stars: Inferring Hidden Fan Support Under Voting-Rule Constraints

Dancing with the Stars combines expert evaluation with audience voting, but the fan-vote component is secret. This makes Problem C fundamentally different from a direct prediction problem: the observed eliminations do not reveal a unique set of fan votes. Our solution therefore treats fan support as a latent variable. Instead of pretending to know exact vote totals, we reconstruct weekly competitions, translate observed eliminations into feasible fan-support constraints, estimate maximum-entropy fan-share distributions, and use those estimates to compare voting rules and propose a producer-facing rule.

Using the official data set of 421 contestants across 34 seasons, we rebuilt 2,777 active contestant-week records and modeled 261 elimination weeks. The resulting latent-support model exactly reproduced the observed modeled eliminations by construction and confirmed that the underlying fan-vote uncertainty is large. This non-identifiability is the central practical finding: many fan-support profiles can explain the same observed elimination. It means that a good voting system should not merely chase an estimated “true” fan vote, but should be robust to close and ambiguous cases.

Our counterfactual simulation shows that rank and percent rules disagree in 26.4% of modeled elimination weeks, while estimated fan support changes the judge-only bottom set in 63.6% of modeled weeks. We also examine controversial contestants, professional-dancer and celebrity-category effects, and the institutional role of bottom-two judge saves. These results support a hybrid recommendation: use a percent-based combined score for transparency, but introduce a gray-zone judge save when the bottom-two combined margin is small. In our simulation, the gray-zone trigger activates in 61.7% of modeled elimination weeks, making close calls explicit rather than hiding them inside opaque vote totals.

For producers, the recommendation is simple: keep fan influence visible, avoid overclaiming exact fan support, and reserve judge discretion for statistically fragile bottom-two cases. The proposed system is more explainable than a pure rank system and more stable than treating a tiny combined-score gap as decisive.

1 Introduction

Dancing with the Stars (DWTS) is a useful voting-system case study because it combines two sources of evaluation that are deliberately different: judges score dance quality, while the public votes for contestants. The official Problem C data set reports contestant information, final outcomes, placements, and weekly judges' scores, but it does not reveal the fan votes. The hidden vote component creates the central modeling difficulty: a contestant can be eliminated under many different fan-vote profiles, and the same public outcome does not identify a unique distribution of fan support.

This observation changes the problem from a direct prediction task into an inverse problem. A supervised model that “predicts” fan votes would be unverifiable because no fan-vote labels exist in the data. Instead, we use the known voting systems as constraints. For each week, the observed elimination is treated as evidence that the true fan support must have belonged to a feasible set. We then estimate a conservative point inside that set and use the uncertainty itself as part of the answer.

Our approach has four linked layers. First, we reconstruct each weekly competition from the wide official data. Second, we formalize the rank, percent, and bottom-two judge-save mechanisms. Third, we estimate latent fan support using feasible-set constraints and a maximum-entropy principle. Fourth, we use the estimated support profiles for counterfactual simulations and a producer-facing voting recommendation. Figure 1 summarizes the complete workflow.

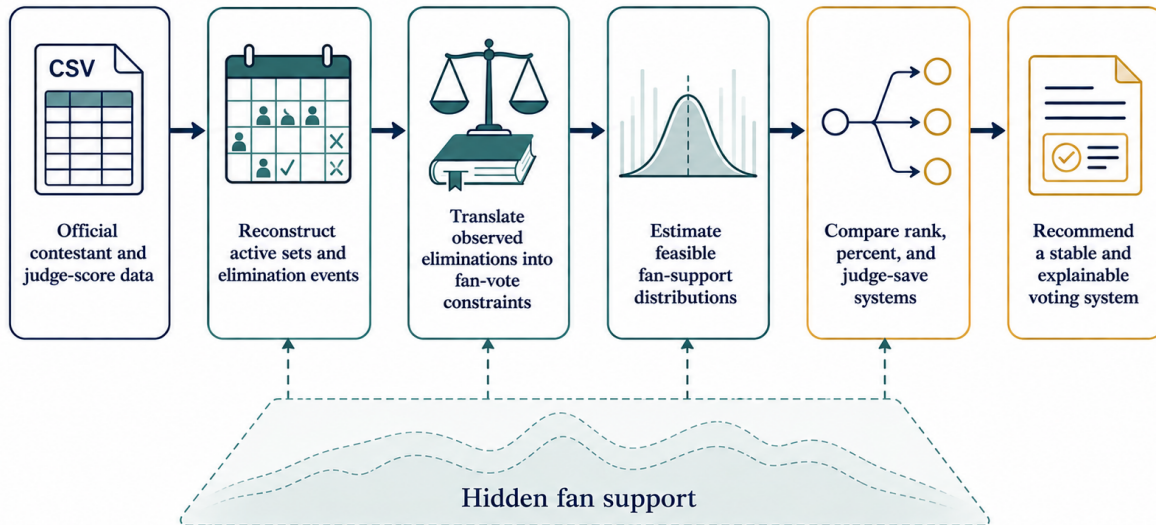


Figure 1: Overall research workflow. Fan support is treated as a hidden variable constrained by observed eliminations rather than as an observed training label.

The main contribution of the paper is not a claim that exact fan votes can be recovered. Rather, we show how to quantify what can be inferred from the publicly observable record, how different voting rules behave under the same latent support estimates, and how producers can design a rule that remains

transparent when the bottom of the leaderboard is statistically fragile.

2 Data Audit and Preprocessing

2.1 Raw data audit

The official CSV contains 421 contestant records, 53 columns, and 34 seasons. For each contestant, the data include identity variables, celebrity industry, home location, age during season, final result, placement, and four possible judge-score columns for each week from Week 1 to Week 11. The raw file is a wide table: one row is one contestant, while weekly scores appear in repeated columns. This layout is convenient for storage but not for modeling a weekly elimination system.

We therefore first audited the data before fitting any model. Missing score cells can have several meanings: a fourth judge may not exist in a given week, a season may not include that week, or a contestant may already have been eliminated. In contrast, explicit zeros after a contestant leaves the competition should not be interpreted as poor performance by an active contestant. This distinction is important because the fan-vote constraints are constructed week by week from the active set.

Table 1: Core data audit metrics used by the modeling workflow.

Metric	Value
Contestant records	421
Raw columns	53
Seasons	34
Reconstructed panel rows	4,631
Active contestant-week rows	2,777
Reconstructed elimination-event rows	343
Single-elimination rows	221
Multi-elimination rows	82
No-elimination rows	40

2.2 Panel reconstruction

We converted the wide table into a long panel indexed by season, week, and contestant. For each contestant-week we computed the available judge count, judge total, judge mean, and an active-status flag. A contestant is active in a week if at least one judge score is present and the weekly judge total is positive. This rule avoids treating post-elimination zeros as active competition records.

Next, we reconstructed the weekly active sets and observed elimination events. Let $A_{s,t}$ be the active set in season s and week t . A contestant in $A_{s,t} \setminus A_{s,t+1}$ is recorded as eliminated after week t . If this set contains multiple contestants, the week is marked as a multi-elimination event. If the active set does not shrink, the week is marked as a no-elimination week. This reconstruction produced 2,777 active contestant-week rows and 261 modeled elimination weeks used by the fan-support inference model.

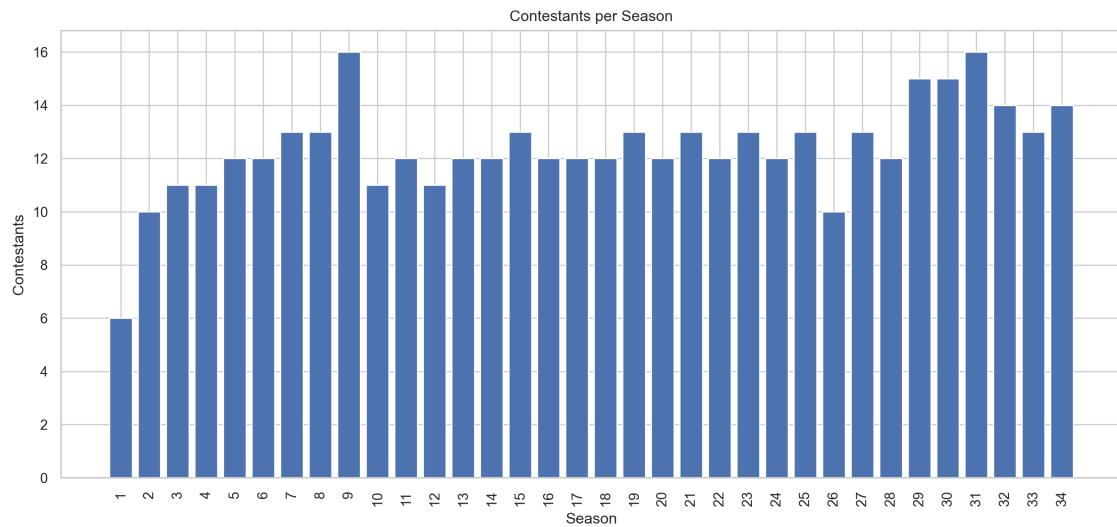


Figure 2: Contestants per season in the reconstructed data set. Season sizes and valid competition weeks vary substantially, so all modeling is done on reconstructed season-week active sets.

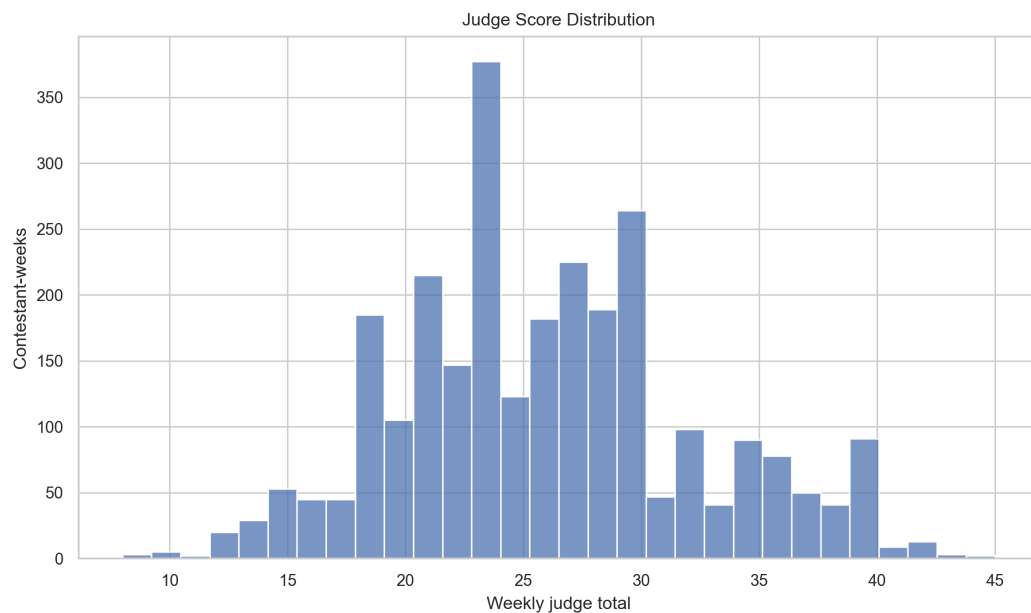


Figure 3: Distribution of active weekly judge totals. These observed scores provide the judge component of each voting rule.

3 Voting Rule Formalization

The contest history involves several ways of combining judges and viewers. We formalize three mechanisms: a rank rule, a percent rule, and a bottom-two judge-save rule. Figure 4 gives the conceptual comparison.

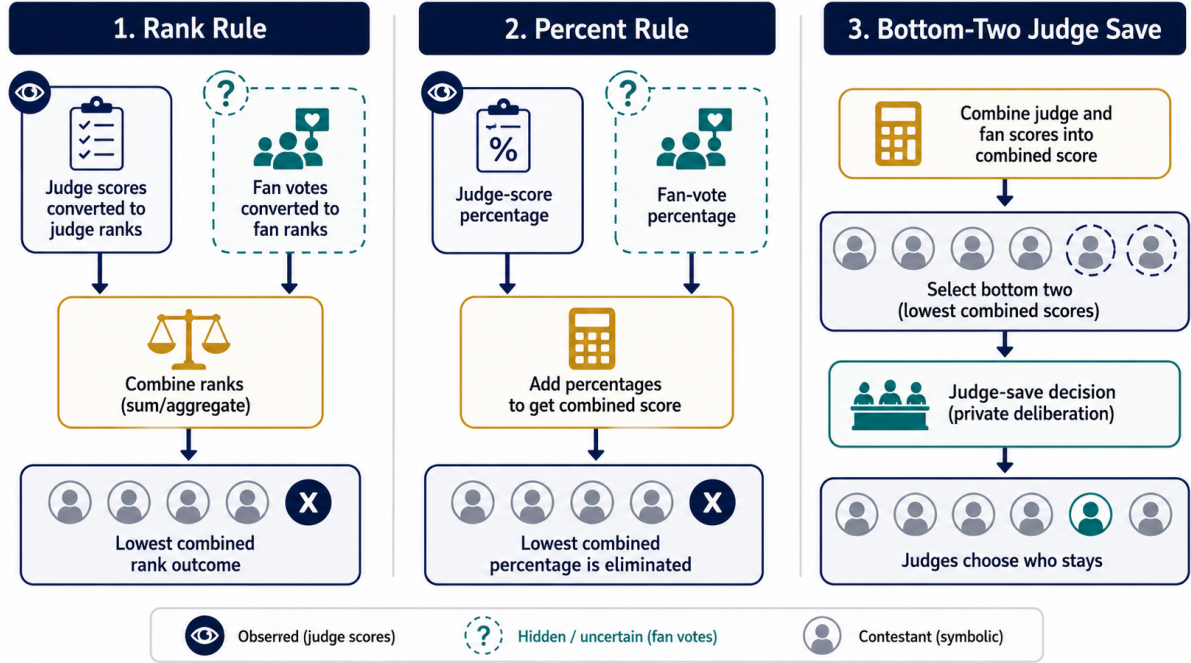


Figure 4: Mechanism comparison for rank, percent, and bottom-two judge-save systems. The fan component is hidden and must be inferred from constraints.

For season s , week t , and active contestant $i \in A_{s,t}$, let J_i be the observed total judge score and let F_i be the unknown fan-share variable. We normalize fan shares so that

$$F_i \geq 0, \quad \sum_{i \in A_{s,t}} F_i = 1.$$

Under a percent rule, judge scores and fan shares are both expressed as shares. The normalized judge share is

$$P_i^J = \frac{J_i}{\sum_{k \in A_{s,t}} J_k},$$

and the combined score is

$$C_i^{\text{pct}} = P_i^J + F_i.$$

The lowest combined score is eliminated. If the observed eliminated set for a week is $E_{s,t}$ and the survivors are $S_{s,t} = A_{s,t} \setminus E_{s,t}$, then every eliminated contestant must have combined score no larger than every survivor:

$$P_e^J + F_e \leq P_j^J + F_j, \quad e \in E_{s,t}, \quad j \in S_{s,t}.$$

These inequalities define a feasible fan-support polytope.

Under a rank rule, the judge score and fan support are converted into ranks:

$$C_i^{\text{rank}} = R_i^J + R_i^F,$$

where larger C_i^{rank} means a worse combined rank. The contestant or contestants with the worst combined rank are eliminated. We use the maximum-entropy fan-share estimate to induce fan ranks for rank-rule counterfactuals.

The bottom-two judge-save mechanism is represented as a two-stage process. First, a combined score identifies the bottom two contestants. Second, judges choose which contestant to save. Because detailed judge-save votes are not fully observed in the data, we use it as a policy simulation rather than a direct historical reconstruction.

Table 2: Voting-rule formalization used in the simulations.

Rule	Score	Elimination decision
Percent	judge share + fan share	lowest combined percent
Rank	judge rank + fan rank	worst combined rank
Bottom-two judge save	combined score, then judge decision	judge save among bottom two

4 Fan Vote Estimation

4.1 Why fan votes are not directly predictable

The fan vote is hidden and non-identifiable from the observed data alone. If a contestant is eliminated, we learn that the combined judge-fan result placed that contestant at or near the bottom under the relevant rule. We do not learn the exact vote total, the absolute number of votes, or the unique share of each contestant. This is why our model estimates feasible support rather than exact votes.

4.2 Feasible fan-support intervals

For each modeled elimination week, we build the percent-rule linear constraints from the observed eliminated set. We then solve two linear programs for each contestant i :

$$\min F_i \quad \text{and} \quad \max F_i$$

subject to the simplex constraints and the elimination inequalities above. The result is an interval $[F_i, \bar{F}_i]$ that describes how much fan support contestant i could have had while still being consistent with the observed elimination. We used SciPy's linear programming solver for this step [6].

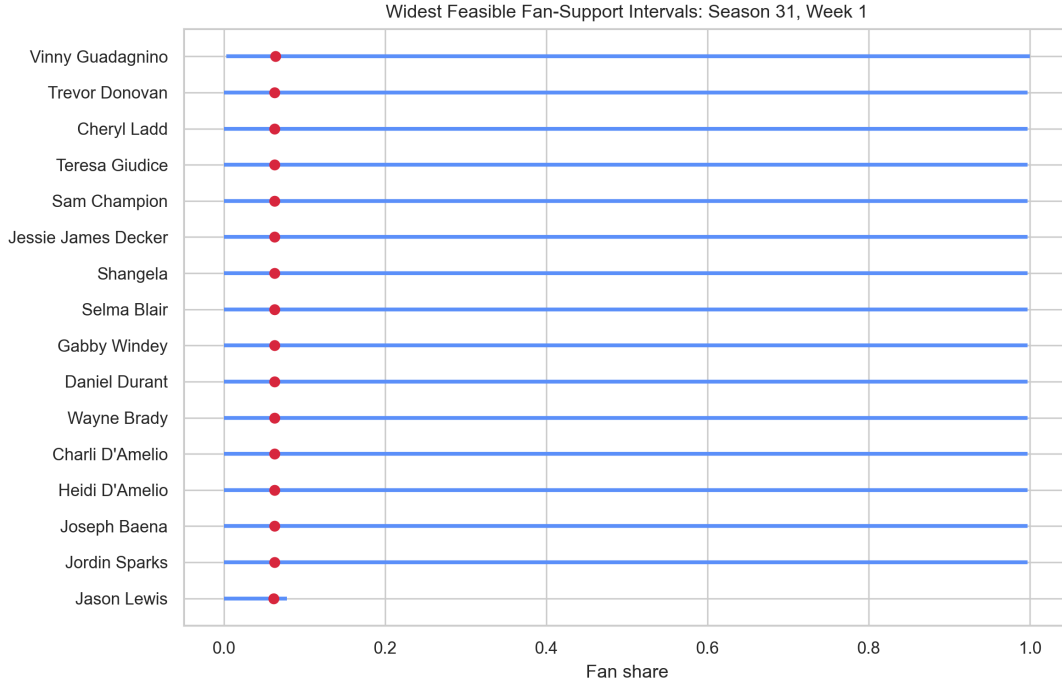


Figure 5: Example of feasible fan-support intervals for a high-uncertainty season-week. The red points show maximum-entropy estimates inside wide feasible intervals.

4.3 Maximum-entropy point estimate

Intervals are useful for uncertainty, but counterfactual simulations also need a single support profile. We choose the maximum-entropy feasible profile:

$$\max_F - \sum_{i \in A_{s,t}} F_i \log F_i$$

subject to the feasible-set constraints. The entropy objective selects the least concentrated fan-support vector consistent with the observed outcome. It therefore avoids arbitrarily inventing a dramatic fan favorite unless the elimination constraints require one. This follows the maximum-entropy principle introduced by Shannon [3] and is implemented as a convex optimization problem using CVXPY [5].

4.4 Uncertainty as a model output

The average feasible interval width across modeled weeks is 0.862, which is large on the $[0, 1]$ fan-share scale. This does not mean the model is weak; it means the historical outcomes do not uniquely identify fan votes. That uncertainty is itself a substantive result. It warns against overinterpreting any one estimated fan-support profile and motivates a voting rule that handles close bottom-two cases explicitly.

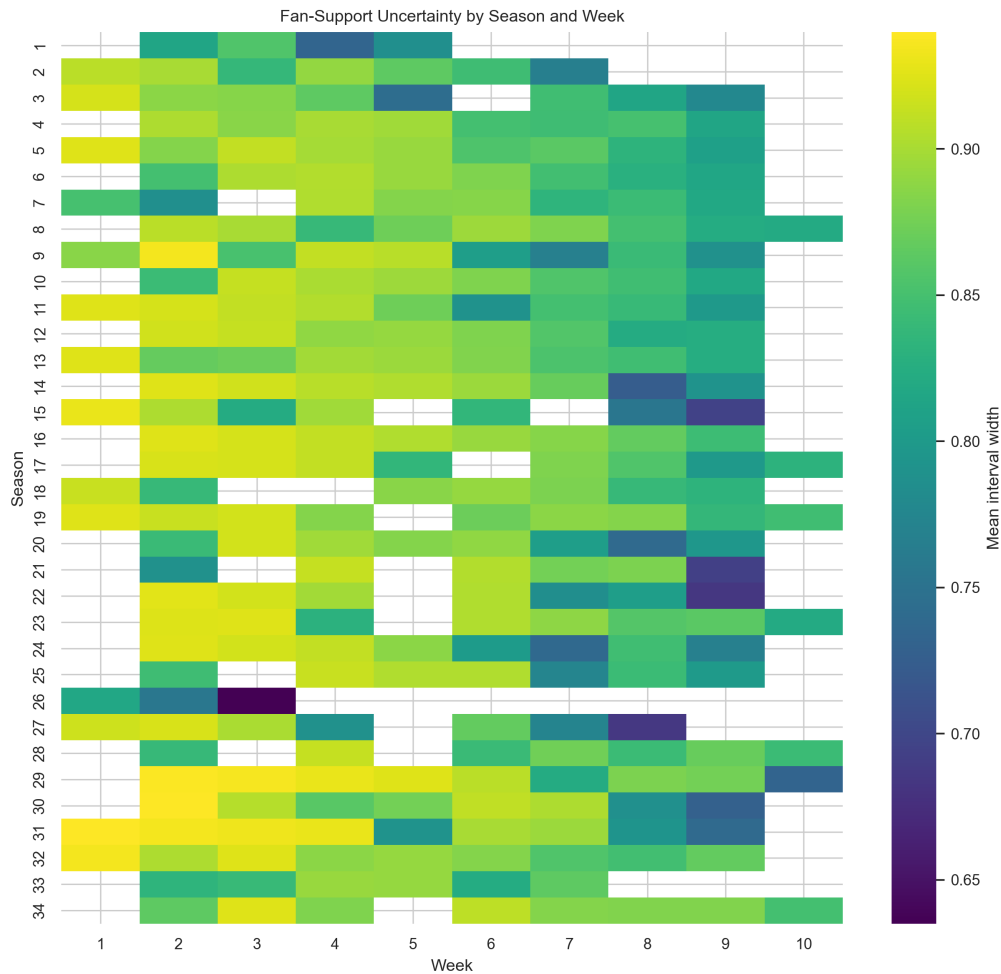


Figure 6: Fan-support uncertainty by season and week. Wide feasible intervals indicate that many latent fan-vote profiles can produce the same observed elimination.

5 Model Validation and Consistency

The validation step asks a limited but essential question: after reconstructing the active set and solving for a feasible maximum-entropy support vector, does the model reproduce the observed elimination under the percent-rule constraints? The answer is yes for all 261 modeled elimination weeks. This 100% reproduction rate should not be misread as out-of-sample prediction. It is a consistency check showing that the reconstructed data, constraints, and optimization pipeline agree with the observed historical record.

Table 3: Validation and uncertainty metrics.

Metric	Value
Modeled elimination weeks	261
Percent-rule reproduction rate	100.0%
Observed elimination in modeled bottom two	100.0%
Mean feasible fan-share interval width	0.862

The more important validation result is the scale of uncertainty. The average interval width of 0.862 shows that observed eliminations alone leave the fan support profile highly underdetermined. In practical terms, a reported elimination may be consistent with a wide range of public-vote configurations. This strengthens the case for comparing voting systems through robust counterfactual behavior rather than through a single supposed true vote count.

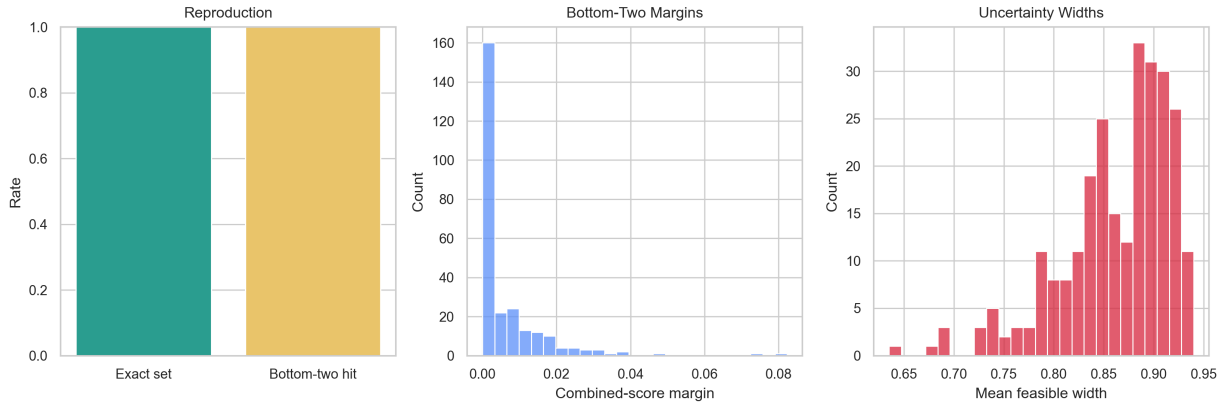


Figure 7: Validation dashboard. The model reproduces the observed modeled eliminations, while the margin and interval-width panels show that close and uncertain weeks are common.

6 Rank vs. Percent Counterfactual Analysis

We next use the same maximum-entropy fan-support estimates to compare the rank and percent rules. This isolates the effect of the rule itself: the contestants, judge scores, and estimated fan support remain fixed, while only the rule used to combine them changes.

Across 261 modeled elimination weeks, rank and percent rules produce different elimination sets in 26.4% of weeks. Estimated fan support changes the judge-only bottom set in 63.6% of weeks. These two metrics have different interpretations. The first measures institutional sensitivity to rule choice. The second measures how often the public component is strong enough to change what would have happened under judges alone.

Table 4: Counterfactual rule-comparison metrics.

Metric	Value
Modeled elimination weeks	261
Rank-percent outcome difference rate	26.4%
Fan override rate relative to judge-only bottom set	63.6%
Mean rank-percent eliminated-set overlap	0.893

The heatmap in Figure 8 shows that rule disagreement is not confined to a single season. It appears throughout the history of the data, which means that voting-rule design is not a minor technicality. A rule change can alter which contestants are at risk even when the same hidden support estimate is used.

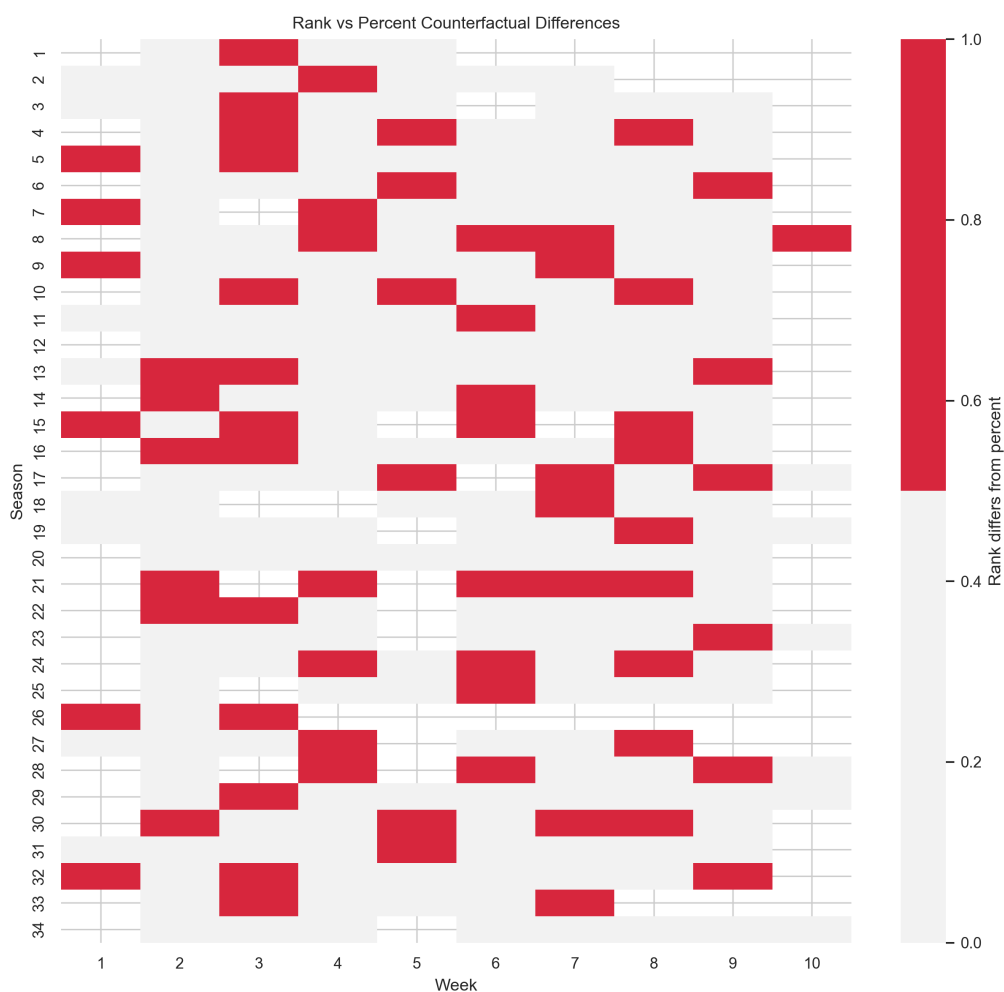


Figure 8: Season-week map of rank-versus-percent differences. Dark cells mark modeled weeks where the two rules select different eliminated sets.

7 Controversial Contestants

The problem statement highlights cases where public popularity and judge scores may appear to pull in different directions. We therefore tracked Jerry Rice, Billy Ray Cyrus, Bristol Palin, and Bobby Bones using a controversy index,

$$CI_{i,t} = |R_{i,t}^J - R_{i,t}^F|,$$

where R^J is judge rank and R^F is the estimated fan rank. Large values indicate a large gap between expert evaluation and inferred public support.

Table 5: Summary of selected controversial contestants.

Contestant	Season	Weeks	Mean judge rank	Mean fan rank	Result
Bristol Palin	15	4	10.50	4.25	Eliminated Week 4
Bobby Bones	27	7	7.29	4.86	1st Place
Bristol Palin	11	9	6.22	3.89	3rd Place
Billy Ray Cyrus	4	7	5.86	4.00	Eliminated Week 8
Jerry Rice	2	7	4.86	4.14	2nd Place

The pattern is clearest for Bristol Palin's Season 15 run and Bobby Bones's Season 27 win. In both cases, estimated fan ranks are substantially better than judge ranks over multiple weeks. This does not prove a specific vote total, but it does quantify the direction of the conflict: these contestants survived longer than judge rank alone would suggest because the feasible fan-support estimates consistently place them higher with viewers.

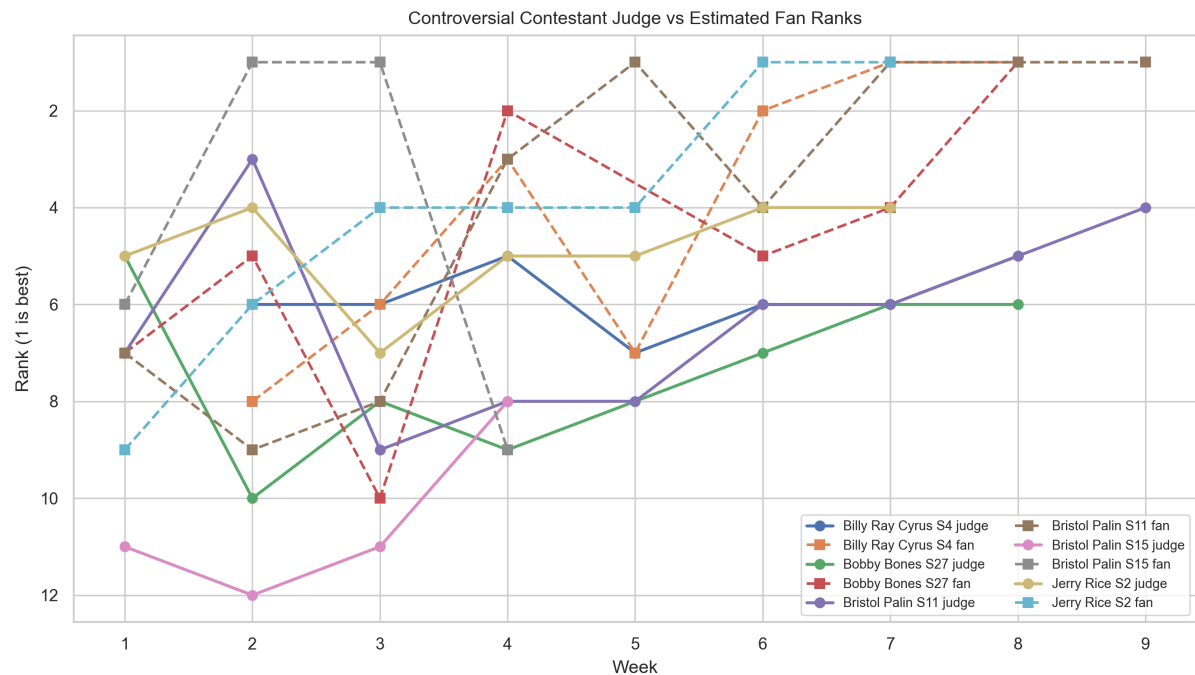


Figure 9: Judge rank versus estimated fan rank for selected controversial contestants. Lower rank values are better.

8 Professional Dancer and Celebrity Effects

We modeled three related effects: judge-score effects, fan-support effects, and survival-time effects. Judge and fan effects were estimated with ordinary least squares using celebrity age and industry indicators. Survival effects were estimated with a penalized Cox model using active-week duration and an elimination indicator. These models are exploratory rather than causal: the data are observational and contestants are not randomly assigned to industries or professional partners.

The strongest consistent result is that age is associated with weaker estimated fan support and shorter survival. In the fan-support model, celebrity age has a negative coefficient of -0.00051 per year. In the survival model, age has a positive hazard coefficient of 0.0327 , meaning older contestants tend to face earlier elimination after controlling for the included industry indicators. Some industry coefficients are also notable: for example, radio personalities and politicians receive lower judge-score estimates in the judge-score model, while social media personalities and TV personalities have lower estimated fan-share coefficients in the fan-support model. These findings should be interpreted as association patterns, not as causal claims.

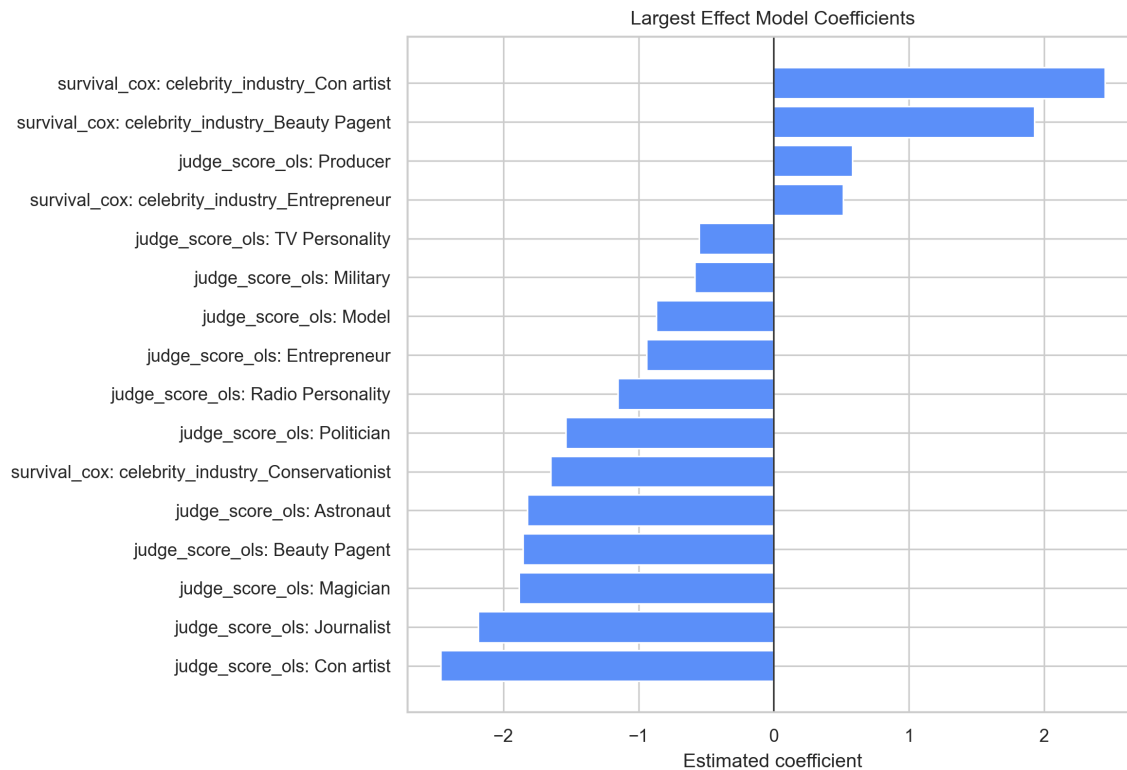


Figure 10: Largest coefficients from the exploratory judge-score, fan-support, and survival models. Coefficients summarize associations, not causal effects.

9 Proposed Voting System

The central design lesson is that close bottom-two cases should be treated as ambiguous rather than forced into a deterministic public-facing ranking. We therefore propose a hybrid system:

Percent Rule + Gray-Zone Judge Save.

The system first converts judge scores and fan votes into percentages and adds them. If the combined-score gap between the lowest two contestants is larger than a threshold τ , the lowest combined score is eliminated. If the margin is at or below τ , the bottom two enter a judge-save decision. In our simulation, τ is set to 0.00354, the 25th percentile of positive modeled bottom-two combined-score margins.

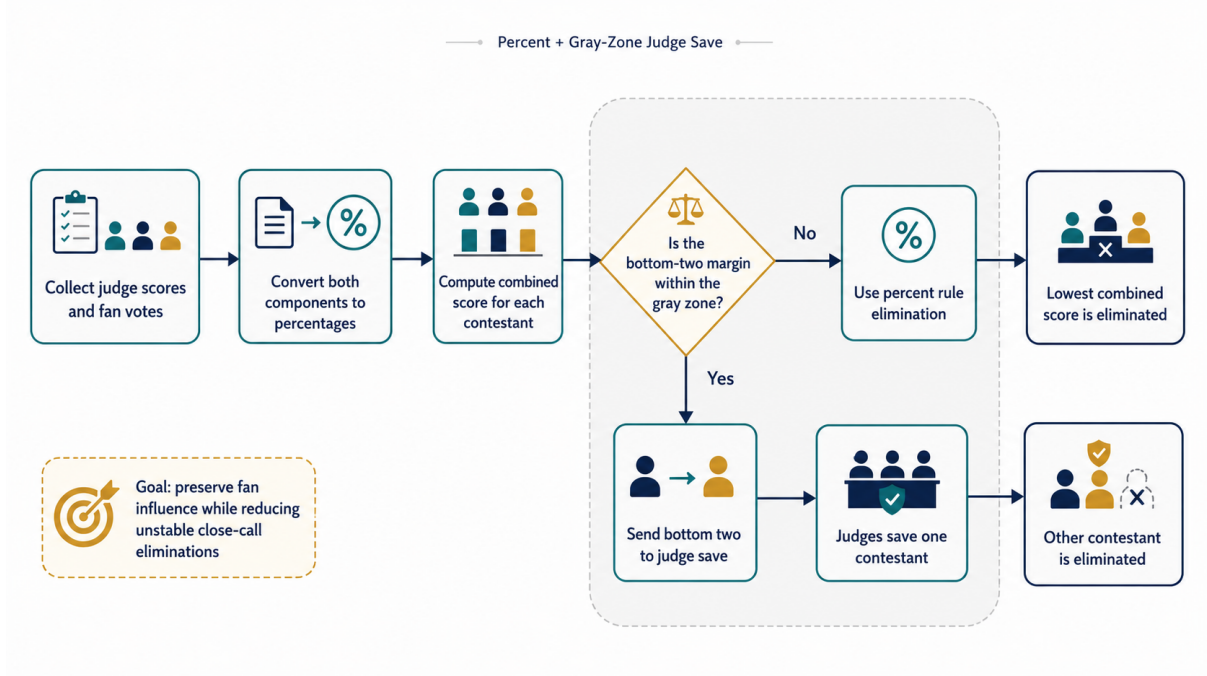


Figure 11: Proposed Percent + Gray-Zone Judge Save system. The judge save is activated only when the bottom-two combined-score margin is small.

The proposed system has three advantages. First, percent scoring is more transparent than rank aggregation because judge and fan shares are measured on the same scale. Second, the gray zone acknowledges that tiny margins are not substantively robust when fan support is hidden and highly non-identifiable. Third, judge discretion is limited to close bottom-two cases, which preserves fan influence while giving producers a defensible way to manage fragile eliminations.

Table 6: Simulation summary for the proposed gray-zone system.

Metric	Value
Gray-zone threshold τ	0.00354
Gray-zone trigger rate	61.7%
Proposed system differs from pure percent rule	46.0%
Proposed elimination appears in observed set	54.0%

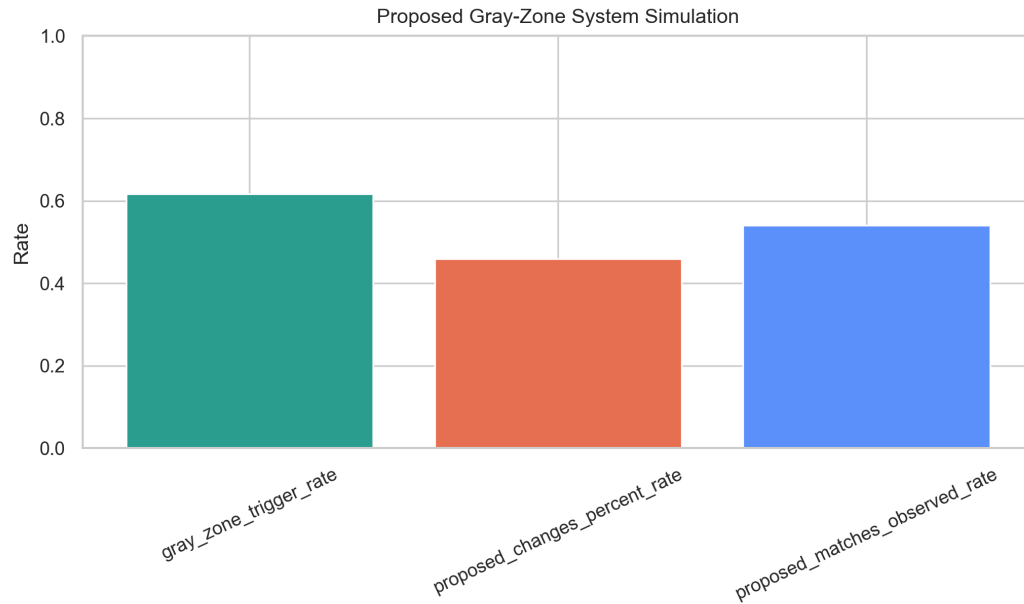


Figure 12: Simulation dashboard for the proposed gray-zone system.

10 Memo to DWTS Producers

To: DWTS Producers

Subject: A transparent voting system for close eliminations

The analysis suggests that fan votes should remain central to the show, but close bottom-two cases should not be treated as if the hidden fan vote were fully observable. We recommend replacing opaque rank aggregation with a Percent + Gray-Zone Judge Save system.

- Convert judge scores and fan votes into percentages before combining them. This makes the rule easier to explain to viewers.
- Use the standard percent result when the bottom-two margin is clearly separated.
- When the bottom-two margin is inside the gray zone, ask judges to save one contestant from the bottom two.
- Publicly describe the judge-save condition as a close-call safeguard, not as a replacement for fan influence.

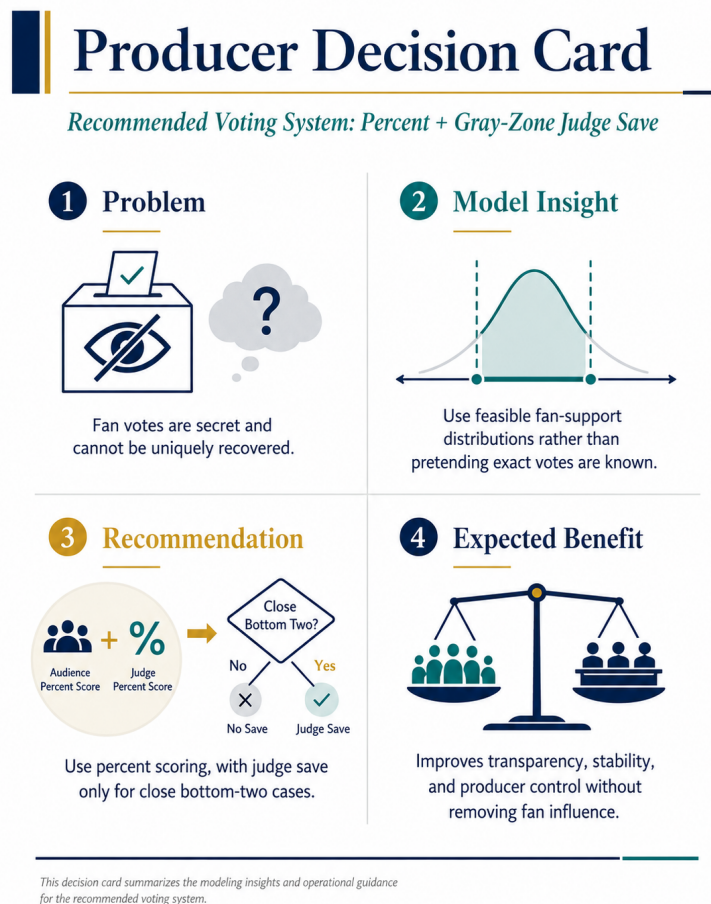


Figure 13: Producer decision card summarizing the recommended voting system.

11 Strengths, Limitations, and Extensions

Strengths. The model respects the fact that fan votes are hidden. It does not invent labels or claim to recover exact vote totals. Every numerical result in the paper is generated from the same reproducible workflow, and the validation step confirms that the reconstructed constraints are internally consistent with the observed modeled eliminations.

Limitations. The maximum-entropy fan profile is a conservative point estimate, not a direct observation. The bottom-two judge-save simulation is a policy model because detailed historical judge-save ballots are not fully available in the data. The professional-dancer and celebrity-effect regressions are observational associations and may reflect unmodeled casting, editing, or audience factors.

Extensions. Future work could add verified external popularity signals, such as social media engagement or search interest, but only if those sources are logged and validated. Another extension would model the evolution of fan support across weeks with a state-space model. We did not use those extensions in the main workflow because the official data already support a complete and transparent constraint-based solution.

References

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